

Optimization of Joint Equalization of High-Speed Signals using Bayesian Machine Learning

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Abstract—The paper elaborates an efficient algorithm for optimization of joint Feed-Forward Equalization (FFE) and Decision Feedback Equalization (DFE) for non-return-to-zero (NRZ) and 4 level pulse amplitude modulation (PAM-4) signals using Bayesian Machine Learning approach previously introduced for NRZ by authors and expanded for PAM-4. A new optimal covariant function and hyper-parameters has been selected for the Bayesian optimization. Cost function for the Bayesian optimization is chosen based on eye height. The proposed method was compared to the conventional Least Mean Square (LMS) method and showed significant improvement. Test cases were performed for several data rates of NRZ and PAM-4 signals with crosstalk and injected jitter. Test results show that the proposed algorithm is the more effective the higher data rates are considered.

Keywords—Equalization; DFE; FFE; Tap coefficients; Joint optimization; Bayesian Optimization; Covariant Function, Cost Function

I. INTRODUCTION

Demand for higher data rates is on a constant increase due to fast development of electronic equipment transmitting signal. Achieving higher signal transmission speed is possible by improving materials and structures of electronic components that is related to higher costs, but also by introducing novel approaches to signal transmission, as well as improving existing algorithms of signal equalization in legacy components. Improving equalization algorithms would be useful for traditional binary NRZ signal and especially for multi-level signal, such as PAM-4, use of which increased during the last several years. The electronic devices consist of many different PCBs, chips, connectors and components. Signal distortion is conditioned by the connection between each component, additional noise, jitter and crosstalk. While such distortions negligibly corrupt low data rate transmission signal and can be

overcome by simpler and straightforward algorithms, their effect increases for the higher data rates and smaller component sizes. Improving the signal requires more sophisticated approach for PAM-4, as its inherent jitter and multi-slicer introduce different complexities. The commonly used equalization methods are Continuous-Time Linear Equalization (CTLE), Feed-forward Equalization (FFE), and Decision Feedback Equalization (DFE) [1] - [9]. The cost function for common equalization optimization algorithms is to lower the difference between the loaded (training) and received (distorted) signals, which is a linear optimization problem and can be solved relatively easily by applying analytical methods. This indirectly will increase the height of the eye diagram. Direct optimization of eye height by using the joint FFE and DFE equalization is a non-linear problem and cannot be solved analytically. For NRZ signals with PRBS-15 (pseudo-random binary sequence) bit sequence, combined FFE and DFE equalization is proposed in [3]. In [5], the goal function minimizes Signal to Noise Ratio (SNR) based on the collaboration of FFE and DFE tap coefficients. The proposed optimization problem is linear regarding FFE and DFE tap coefficients and is solved analytically. Similar to [4], the study in [5] is done for NRZ. The results of [4] was extended for PAM-4, though without taking into account jitter and crosstalk.

In the present paper, the equalization optimization algorithm is applied for NRZ signals with PRBS-15 bit sequence as well as 4 level Pulse Amplitude Modulation (PAM-4) signals with Quaternary PRBS-13 (QPRBS-13) bit sequence. We propose a joint FFE and DFE equalization optimization algorithm that directly optimizes eye height using Bayesian machine learning. By combining the iterative methods and Bayesian machine learning, in this study, the Bayesian Optimization (BO) algorithm finds the best values for tap coefficients of joint FFE

and DFE signal equalization in real-time. The application of BO is one of the popular approaches for many optimization problems in engineering, especially dealing with multi-dimensional cases (see, for example, [10], [11]). Unlike to gradient-based optimization methods, BO does not require information about derivatives. Consequently, it can be used as a powerful derivative-free optimization method for black-box functions. BO is based on Gaussian Process (GP) and Bayes' Theorem (for details see, [12] - [14]). The proposed algorithm is tested for both NRZ and PAM-4 signals for a channel with and without crosstalk and jitter.

The paper is organized as follows. In Section II, some mathematical background for FFE and DFE equalization as well as brief description of BO is given. In Section III, BO is applied for joint FFE and DFE equalization for NRZ (with PRBS-15) and PAM-4 (with QPRBS-13) signals. The results of numerical experiments are done in the same section. The conclusions are drawn in Section IV finally.

II. SOME MATHEMATICAL BACKGROUND

A. Statement of the Joint Equalization Problem

Feed Forward Equalization is a simple convolution of the channel response on FFE tap coefficients. Feed forward equalized signal can be obtained using the following formula:

$$v_k^{FFE} = \sum_{i=-m}^n c_i v_{k-i}, \quad (1)$$

where $c_i, i = -m, \dots, n$, are FFE tap coefficients with m precursor and n post-cursor tap coefficients. c_0 represents a cursor value, $v_k, k = 1, 2, \dots, N$ is a channel response and v_k^{FFE} is equalized signal. The main goal is to find such tap coefficients that allows to recover input signal as accurately as it is possible. Usually, for better improvement of the attenuated signal, after FFE equalization, DFE equalization is applied. If DFE tap coefficients are known, combined FFE/DFE sequential equalization for NRZ signal is given by the following equation:

$$v_k^{FFE/DFE} = v_k^{FFE} - \sum_{i=1}^p d_i \text{sign}(v_{k-i}^{FFE}), \quad (2)$$

where $d_i, i = 1, \dots, p$, are DFE tap coefficients. In (2), function $\text{sign}(x)$ is a discontinuous, non-linear function which equals to 1 for positive x and -1 for negative x . Pictorial description of FFE and combined FFE/DFE sequential equalization are given in Fig. 1 and Fig. 2 correspondingly.

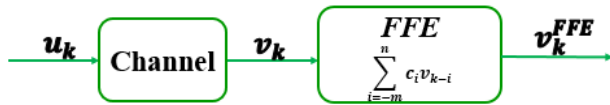


Fig 1. Linear Feed-forward Equalization



Fig 2. Combined FFE/DFE sequential equalization scheme

Purpose of the paper is to find optimal tap coefficients for FFE and DFE equalization that allows to achieve the best Eye parameters after applying joint FFE/DFE equalization to the distorted channel response. Pictorial representation for joint FFE and DFE equalization done on Fig. 3.



Fig 3. Joint FFE&DFE equalization scheme

To apply the Bayesian optimization method, we need to construct the respective cost function. The cost function directly optimizes eye height. In Fig. 4 the input and output signals are aligned for PAM-4 signal. Considering each Unit Interval (UI) of whole signals, all logical levels ($Logic_{-1}$, $Logic_{-1/3}$, $Logic_{1/3}$, and $Logic_1$) for the output signal are compared to the corresponding input signal levels (-1, -1/3, 1/3, and 1) and the lowest and the highest logical levels are found for top, middle, and bottom eyes.

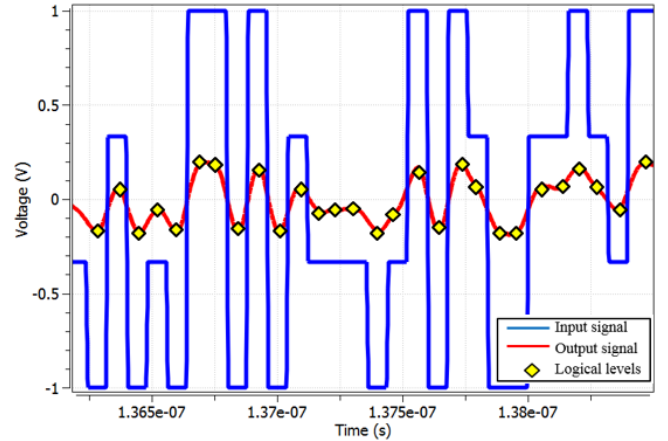


Fig 4. Logical levels for PAM-4

Fig. 5 shows an eye diagrams with eye heights which are calculated as differences between corresponding logical levels for each UI:

$$\begin{aligned} h_{top} &= Logic_1 - Logic_{1/3}, \\ h_{middle} &= Logic_{1/3} - Logic_{-1/3}, \\ h_{bottom} &= Logic_{-1/3} - Logic_{-1}. \end{aligned} \quad (3)$$

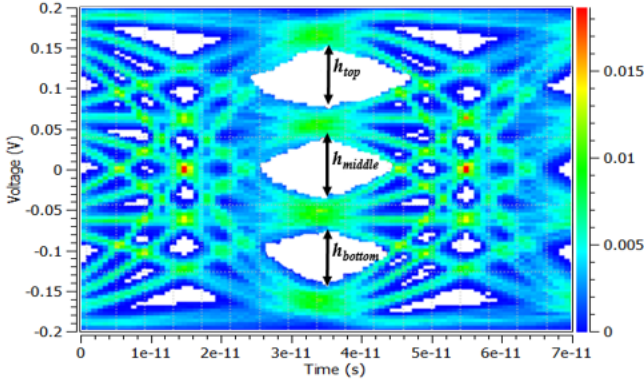


Fig 5. Eye diagram for PAM-4

Cost function incorporates FFE and DFE tap coefficients together and taking into account equalities (3) constructs the output value as follows:

$$h_{PAM4} = \min(h_{top}, h_{middle}, h_{bottom}). \quad (4)$$

Cost function for NRZ signal is constructed similar way, difference is that for NRZ we will have only one eye height.

As we see from equation (4), there is no analytical representation for h_{PAM4} showing explicit dependence on FFE and DFE tap coefficients. Therefore, to maximize eye heights the black-box function optimization method is needed. We applied the Bayesian machine learning for the mentioned black-box optimization. In particular, different covariation functions and hyperparameters have been tried and the optimal one has been selected. A brief description of BO is given below.

B. A short overview of the Bayesian Optimization Algorithm

As we noted, the Bayesian Optimization is based on Gaussian Process and Bayes Theorem [12] - [14]. Whereas a Gaussian probability distribution describes random variables Gaussian process governs the properties of functions [12] - [14]. Although, we avoid the description of GP and BO in this paper, the proposed subsection describes the main steps for the BO algorithm which is given on the flowchart below (Fig. 6) [11]. For details regarding GP and BO, we refer the interested reader to [3], [10] - [16].

The main compound of BO is the covariance function which is applied on the fourth and fifth steps in the flowchart given in Fig. 6. It is a GP predictor, as it encodes the assumptions about the function which is to be studied. One of the most common covariance functions is the Squared Exponential (SE) function, which is defined as follows:

$$SE = k(\mathbf{x}_i, \mathbf{x}_j) = \sigma \exp\left(-\frac{1}{2} \frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\theta^2}\right), \quad (5)$$

where σ and θ are tunable hyper-parameters. The covariance function sets covariance between the outputs and inputs. Note that, covariance function (5) shows that the covariance is almost unity between variables whose corresponding inputs are very close, and decreases as their distance in the inputs increases. It is very important to

understand the similarity between data points in supervised learning; the basic similarity assumption is that two points $\mathbf{x}_i$ and $\mathbf{x}_j$ which are close are likely to have similar outputs, and therefore training points that are close to a test point should have information about the studied function prediction at that point. Under the Gaussian process view, it is the covariance function that defines nearness or similarity (for details see [13]).

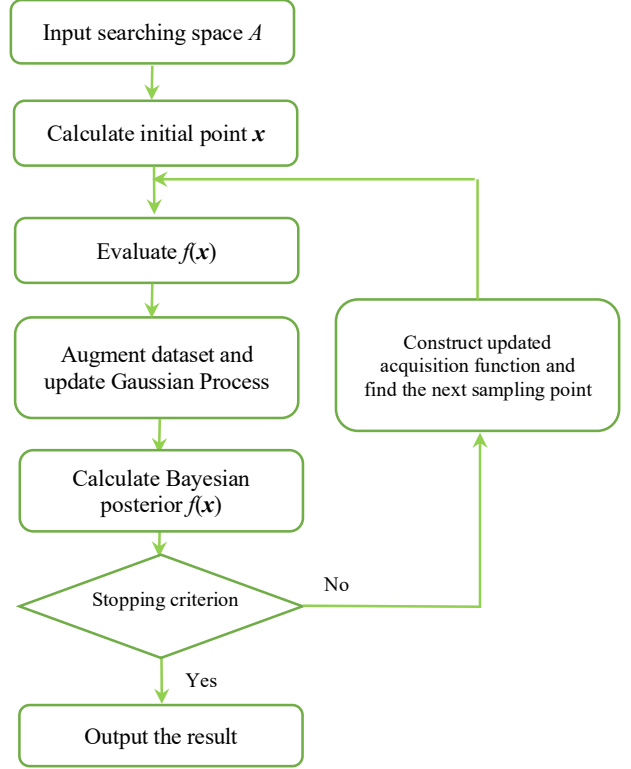


Fig 6. A flowchart for the regular BO algorithm

III. RESULTS OF NUMERICAL EXPERIMENTS

In [3] a high-speed channel is considered to compare BO based equalization against Least Mean Square (LMS) equalization. Eye opening is compared for NRZ and PAM-4 signal with 50 and 56 Gbps data rate. Fig. 9 shows insertion loss, return loss and crosstalk of the channel. The channel comprises of 5" transmission line on a board with Megtron-6 material (ultra-low loss material) connecting to another board with 13" of transmission line on Megtron-6. There are total of 4 vias in the path with each via having less than 10 mils of via stub. The connector used for connecting the boards is a high frequency connector designed for 50 Gbps speeds [3].

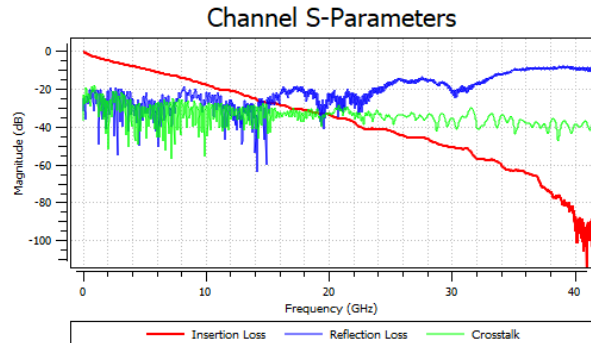


Fig. 7. Insertion loss and return loss of the test channel

Fig. 8a, and Fig. 8b shows eye diagrams for recovered 50Gbps NRZ signals by proposed methodology and conventional LMS methods for the described channel without jitter and crosstalk.

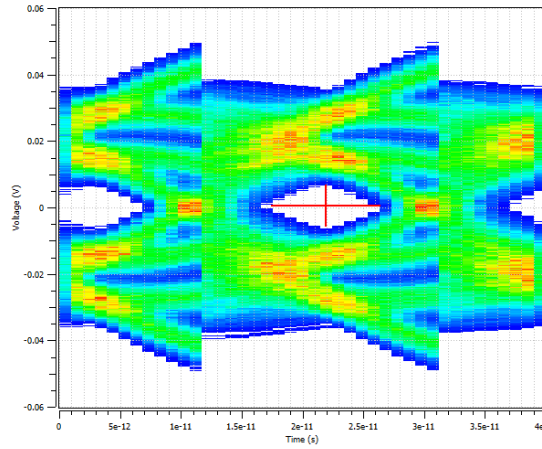


Fig. 8a. Eye Diagram for 50Gbps NRZ signal without crosstalk and jitter using LMS optimization

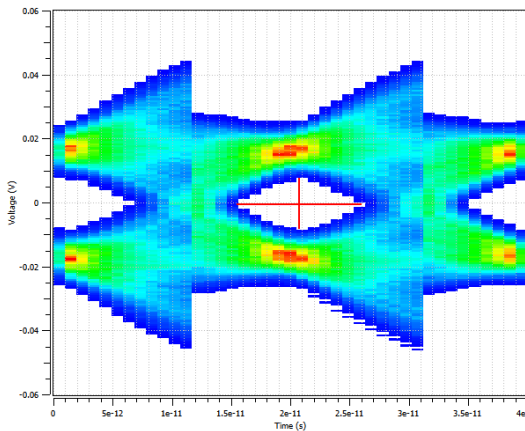


Fig. 8b. Eye Diagram for 50Gbps NRZ signal without crosstalk and jitter using the Bayesian optimization

Eye height for LMS method (See Fig 8a) equals to 12mV and for the Bayesian method (See Fig 8b) equals to 16mV. As we see the Bayesian optimization was able to improve the LSM results by 33%. Besides, as it can be observed from Fig 8a and

Fig 8b signal to noise ratio is much better for the Bayesian optimization.

Fig. 8a, and Fig. 8b shows eye diagram for recovered 50Gbps PAM-4 signal by proposed methodology. Eye height for Bayesian method equals to 29mV. LMS was not able to open eye for PAM-4 at all.

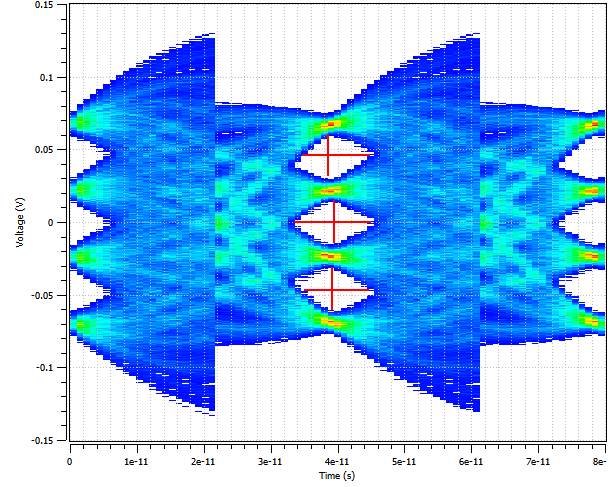


Fig. 9. Eye Diagram for 50Gbps PAM-4 signal without crosstalk and jitter using Bayesian optimization

Fig. 10 shows eye diagram for recovered 50Gbps PAM-4 signal with injected jitter using proposed methodology. Eye height for Bayesian method equals to 12mV. LMS was not able to open eye for the signal with injected jitter at all.

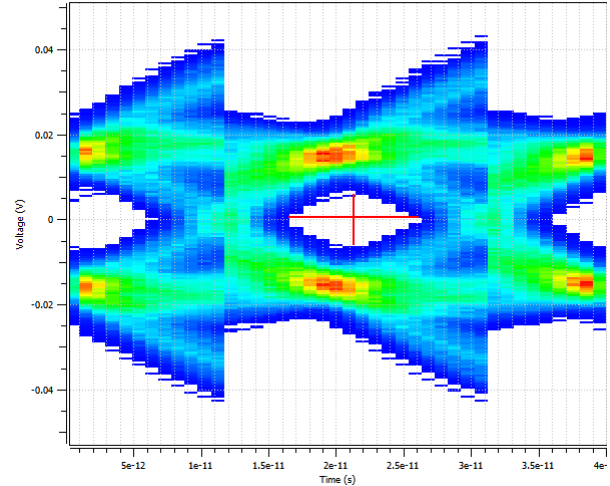


Fig. 10. Eye Diagram for 50Gbps NRZ signal with injected jitter using the Bayesian optimization

Fig. 11 shows eye diagram for recovered 50Gbps PAM-4 signal with channel crosstalk using proposed methodology. Eye height for the Bayesian method equals to 21mV. LMS was not able to open eye for the channel included crosstalk at all

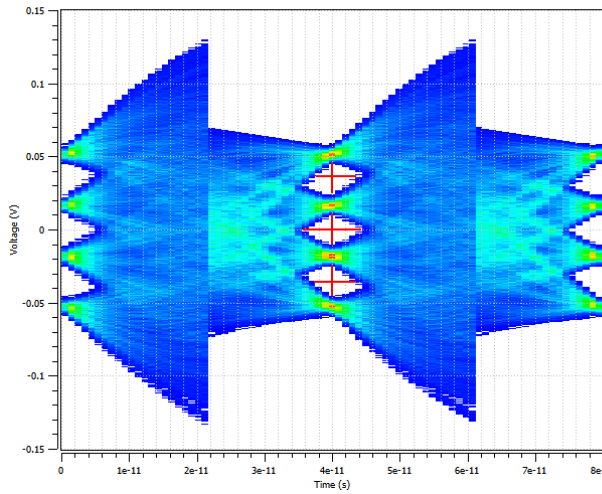


Fig. 11. Eye Diagram for 50Gbps PAM-4 signal with crosstalk using the Bayesian optimization

The following table summarizes eye height and eye width test results for different configuration for both LMS and the proposed Bayesian optimization method. As we can see from Table 1, the proposed Bayesian optimization method gives significant improvement from standard LMS method.

Configuration	Signal Type	Datarate (Gbps)	Eye Height (mV)		Eye Width (ps)	
			LSM	Bayessian	LSM	Bayessian
No Xtk and Jitter	NRZ	50	12	16	9	11
No Xtk and Jitter	PAM4	50	25	29	9	10
No Xtk and Jitter	NRZ	56	0	15	0	11
No Xtk and Jitter	PAM4	56	14	24	6	9
Injected Jitter	NRZ	50	0	12	0	10
Injected Jitter	PAM4	50	18	23	7	7
Injected Jitter	NRZ	56	0	8	0	7
Injected Jitter	PAM4	56	0	16	0	6
With Crosstalk	PAM4	50	0	21	0	6

Table 1. Summarized test results for LMS and the proposed Bayesian optimization methods

IV. CONCLUSIONS

The Bayesian optimization algorithm is used to recover attenuated channel response using joint FFE and DFE equalization for NRZ and PAM-4 signals. Optimal covariant function and hyper-parameters have been selected for the Bayesian optimization. The proposed method was tested for the channel that comprises of 5" transmission line on a board with Megtron-6 material (ultra-low loss material) connecting to another board with 13" of transmission line on Megtron-6. There are total of 4 vias in the path with each via having less than 10 mils of via stub. Tests were performed using crosstalk and injected jitter. The results were compared to standard LMS algorithm for FFE and DFE. The proposed method shows significant improvement of Eye parameters.

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